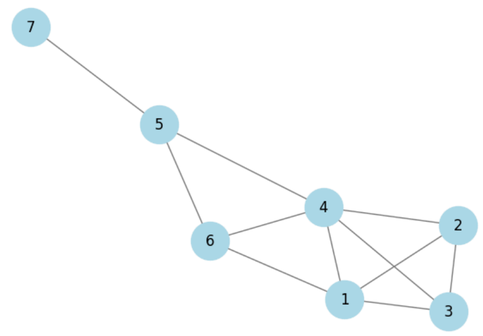
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Module 1 Assignment
Graded Assignment • 30 min

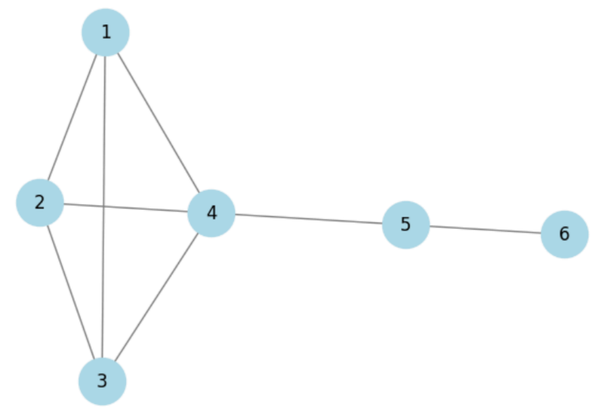
Due Oct 16, 11:59 PM IST

1. Refer to the graph below. After performing a K-core decomposition, which nodes remain in the 3-core? 1 point



- [1, 2, 3, 4]
- [1,2,3,4,5, 6]
- [1,2,3,4,5,6,7]
- [1,2,3,4,5]

2. What is the effective size of node 1 in the network below? 1 point



- 3.0
- 1.0
- 4.667
- 2.0

3. What happens as you increase the value of k in k-core decomposition? 1 point

- The k-core becomes larger and includes more nodes
- The network forms edges between disconnected communities and nodes
- The number of nodes increases
- The k-core becomes smaller, retaining only highly connected nodes

4. Which of the following is a real-world example of a broker spanning a structural hole? 1 point

- A group of close friends who all know each other and frequently interact
- Diplomats between two warring countries who communicate messages on behalf of their governments
- A social media influencer who interacts only with their followers but not with other influencers
- A densely connected online forum where all members are in constant discussion

5. Which of the following real-world situations would be most appropriate to analyze with numeric assortativity? 1 point

- The likelihood of movie actors forming ties based on the genre of films they have acted in
- The tendency for people with similar income to form connections with each other
- The tendency of people to form friendships based on shared political affiliation
- The preference of employees in a company to connect with others in the same department

6. What aligns closest with the concept of a broker in the context of structural holes? 1 point

- A node that connects two otherwise disconnected groups in a network
- A node with the highest degree centrality in the entire network
- A highly connected node within a densely clustered community
- A node that only forms ties with others that share similar attributes

7. Which of the following real-world examples best showcases the concept of homophily in a social network? 1 point

- People of the same age forming friendships with each other
- A kindergarten activity where the teacher pairs up students at random
- A CEO interviewing a diverse group of candidates for a role
- A university student becoming friends with everyone they meet

8. Which of the following real-world situations would be most appropriate to analyze with attribute assortativity? 1 point

- The connections between professors in a collaboration network based on the number of authored papers
- The likelihood of athletes forming social ties based on their years of experience
- The tendency of high school students to form friendships with others that do the same sport
- The tendency for people with similar income to form connections with each other

9. How is k-core decomposition useful in network analysis? 1 point

- It identifies the most influential edges in the network by using the constraint coefficients of nodes connected by the edges to determine whether the edges span structural holes.
- It identifies cohesive subgroups of central nodes that remain connected even when peripheral nodes have been removed
- It identifies the most influential brokers in a network by locating nodes that connect groups of otherwise disconnected communities
- It identifies structural holes in the network

10. What aligns closest with the concept of a structural hole in a network? 1 point

- A missing connection between two communities
- A cycle that allows information to spread efficiently
- A fully connected subgraph within a network
- A node with the highest degree in a network